



Scientific Writing Lesson 4: Writing the proposal and introduction section

WRITING YOUR THESIS OUTLINE

NOTHING SAYS "I'M ALMOST DONE" TO YOUR ADVISOR/
SPOUSE/PARENTS LIKE PRETENDING YOU HAVE A PLAN

STEP 1 Aim for a respectable number of chapters:

THESIS OUTLINE

- 1.
- 2.
- 3.
- 4.
5. ↖ chapter #’s
- 6.
- 7.

5 = "That's IT??"
6-7 = "Not bad"
8+ = "Are you crazy??"

STEP 2 Fill in the "freebies":

THESIS OUTLINE

1. INTRODUCTION
2. LIT REVIEW
3. METHODOLOGY
- 4.
- 5.
- 6.
7. CONCLUSIONS

You're half way done!

STEP 3 Make up titles for the "meat" chapters:

2. LIT REVIEW
3. METHODOLOGY
4. (THAT STUFF YOU DID YOUR FIRST YEAR)
5. (STUFF YOU'RE SUPPOSED TO BE DOING NOW)
6. (MAKE STUFF UP)
7. CONCLUSIONS

(It'll be years before you actually have to work on that later chapter, and by then your thesis topic will have changed anyway)

STEP 4 Voilà! You just bought yourself another two years

So, how's your thesis going?
i have an outline!

JORGE CHAM © 2006

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Open 3 scientific articles or theses in your field of study.



They should be primary research texts, not review articles.

Lesson 4: objectives

1. Proposal writing

- Why write a proposal? Structure. Tips.

2. Introduction section of a thesis or research article

- 3 Moves in research introductions
- Establishing a territory: centrality statement
- Establishing a niche: writing a gap/problem statement
- Occupying the niche/addressing the problem
- Hypotheses / Research questions

3. Using participle forms in scientific writing

- Present, past and perfect participles

Assignment 4

What are the purposes of a research proposal?

- Enables the university or funding institution to evaluate your suitability for graduate level research
- Indicates if your research is new and original
- Assesses if financial support for your proposed area of study can be found
- Could identify the right supervisor for you

What should a good proposal do?

A good proposal should:

- clearly define your intended topic
- demonstrate that you understand your research area
- indicate that you have started to develop an original and interesting research question/hypothesis
- show that you understand how to conduct research
- look professional – ensure your proposal is well-structured, clear and legible, and that you have checked for any spelling or grammar errors
- focus on quality rather than quantity

Research proposal: structure

1. **Title** –clear, concise and describe the theme of your proposal
2. **Overview/Abstract** –summary including problem statement, methods and relevance
3. **Background summary/Introduction** –study area: significance, impact and contribution
4. **Aims/Objectives** –including hypothesis and/or research questions
5. **Methods**– essential detail about how you intend to carry out your research. *Use the future tense here!*
6. **Expected Results/Statistical analysis** (if required)
7. **Time plan with milestones** - a schedule of key activities
8. **Bibliography**

Swiss TPH Handbook: guidelines and template for PhD Proposal <https://team.swisstph.ch/s/MapGAgp7QRuslh7icCAG0w>

You could also write sections of your proposal (Introduction, Methods, Abstract) as the assignments for this course, but obviously not for Results or Discussion – unless you write expected results.

Just inform your reviewer about that, as it will influence content, tense use, etc.

15 KEY TIPS FOR WRITING A WINNING GRANT PROPOSAL



BEFORE YOU START WRITING...



LEARN MORE ABOUT THE FUNDING AGENCY

Read their website to understand their goals and how your research will help in achieving these goals



UNDERSTAND YOUR TARGET AUDIENCE

Read up about the reviewing committee and decide which aspects of your subject matter need to be explained in detail



GO THROUGH THE FUNDER'S GUIDELINES FOR PROPOSALS

Make a list of everything you need to do: which questions to address, sections to include, supporting documents to be submitted, formatting specifications to be considered

WHILE WRITING YOUR GRANT PROPOSAL, MAKE SURE YOU TALK ABOUT:

 The usefulness of your research for scientific development and human knowledge	 How the sample used in the study is widely applicable
 What motivated you to choose your research topic	 The expertise and experience of all the researchers who are part of your study
 How your work is aligned with the funding institution's mission and vision	 Details of how your institution is equipped to support you, and the facilities it offers to help your research
 The problems you have identified and the solutions your study will provide	 The resources you need or plan to use to conduct your research
 The credibility of the methods you plan to use	 Details of how your study is in compliance with relevant ethical standards
 How your study has a focused and measurable aim	 How you are equipped to lead your research team well



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Resources throughout your publication journey

Tailor each grant application proposal to the funding agency you are applying for – generic applications will not be successful.

Riana Paola Swiss TPH Scientific Writing

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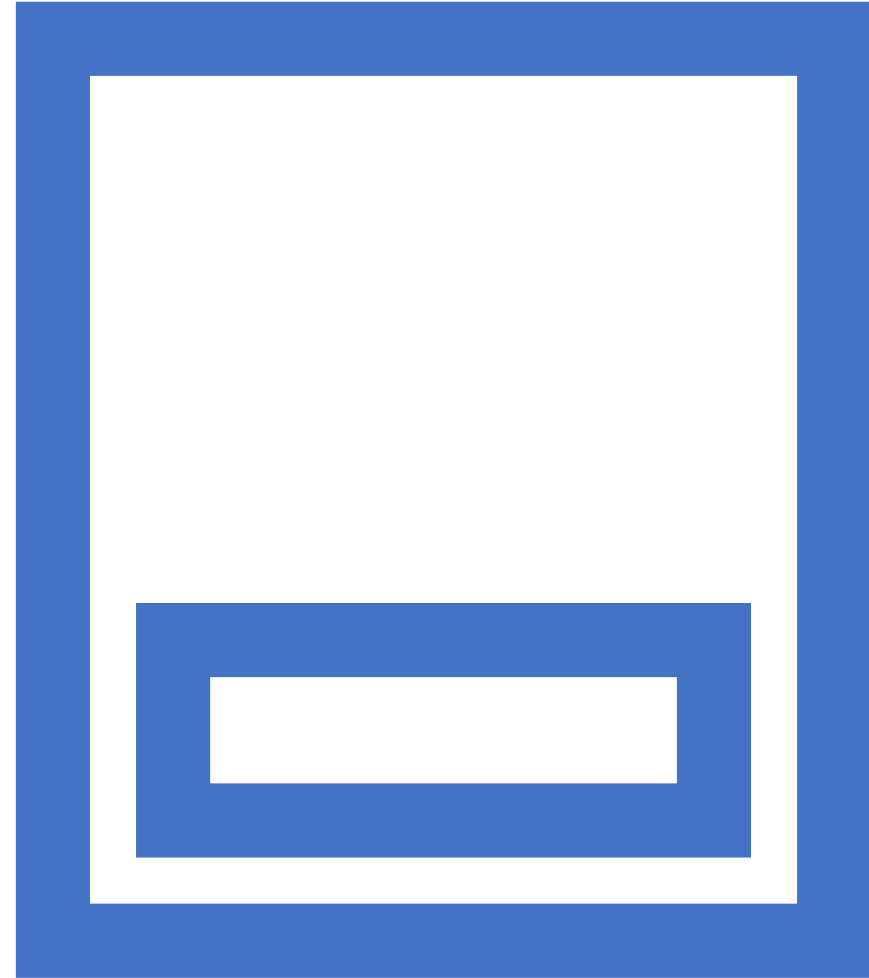
Generic thesis structure: IMRaD

1. Introduction /Background /Literature Review
2. Theory /Approach /Methods and Materials/
Subjects
3. Results /Analysis
4. Interpretation /Discussion
5. Conclusions /Implications
/Recommendations

Exploit the thesis/article as the basis for other publications!

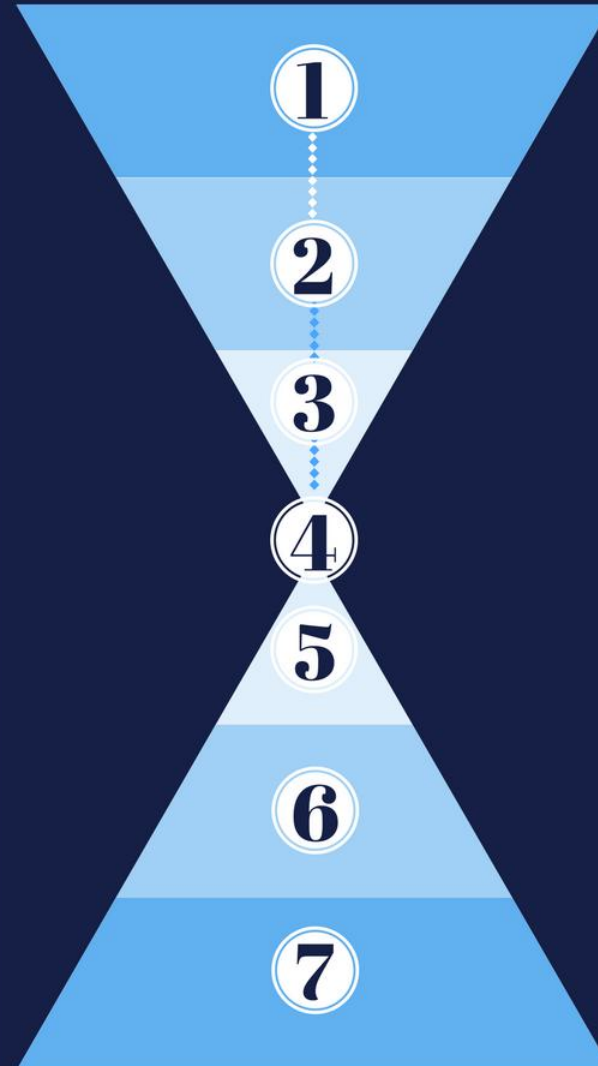
Where do you start?

- What section of your thesis/article are you writing first? Why?
- Do you think there could be other starting points?
- Which section of the thesis/article might be the easiest for you to start with?



In this lesson, we are exploring sections 1-3 of this visual.

THE ANATOMY OF A SCIENTIFIC PAPER



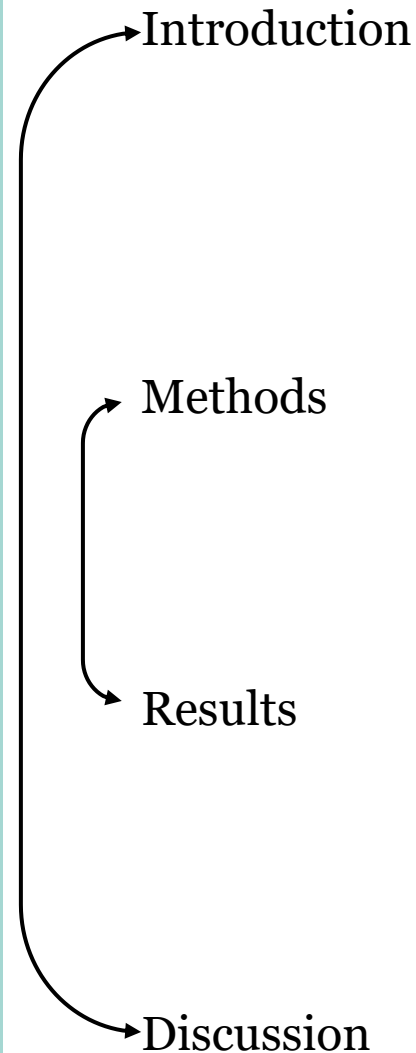
- 1** **INTRODUCTION:**
What is known?
(Our understanding of the world)
- 2** What is unknown?
(What's the gap we want to fill?)
- 3** How and why should we fill the gap?
(Your rationale and purpose/hypothesis)
- 4** **METHODS:**
What did you do?
- 5** **RESULTS:**
What results did you get?
- 6** **DISCUSSION:**
How do the results fill the gap?
- 7** **CONCLUSION:**
What does this mean for us going forward?

Research Article Layout

This is the “**why ?**” section of your manuscript.

In the Introduction, you justify what makes your study necessary/important /a great idea.

From Swales & Feak,
Academic Writing for Graduate Students, 3rd Ed.



General
↓
Specific

WHY?

HOW?

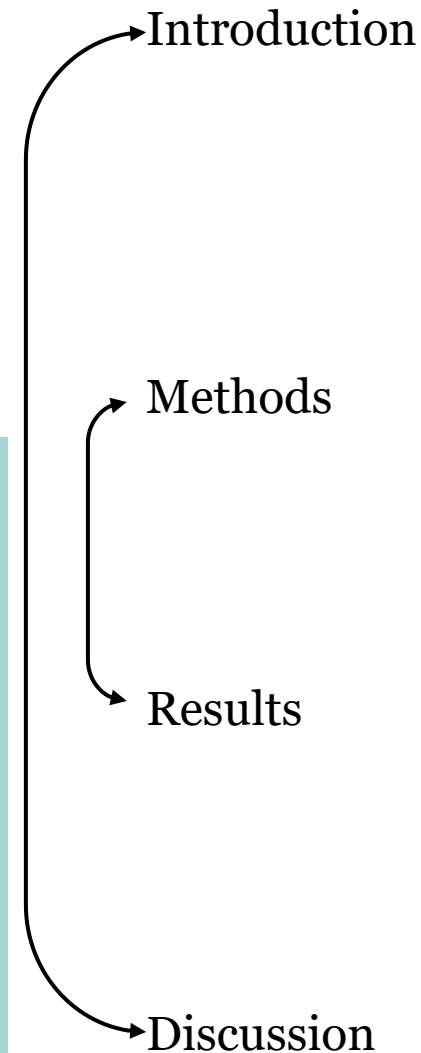
WHAT?

Specific
↓
General

SO WHAT?

Research Article Layout: What to include in each section

From Swales & Feak,
Academic Writing for Graduate Students, 3rd Ed.



Move 1: Centrality Claim

1a: Relevance of the topic

1b: Reviewing previous research

Move 2: Indicating a Gap

Move 3: Filling the Gap

3a: Purpose of research

3b: Research question/hypothesis

3c: Value of research

Subjects / Materials

Location

Procedures

...

Data analysis / Statistics

Move 2: Describe the data

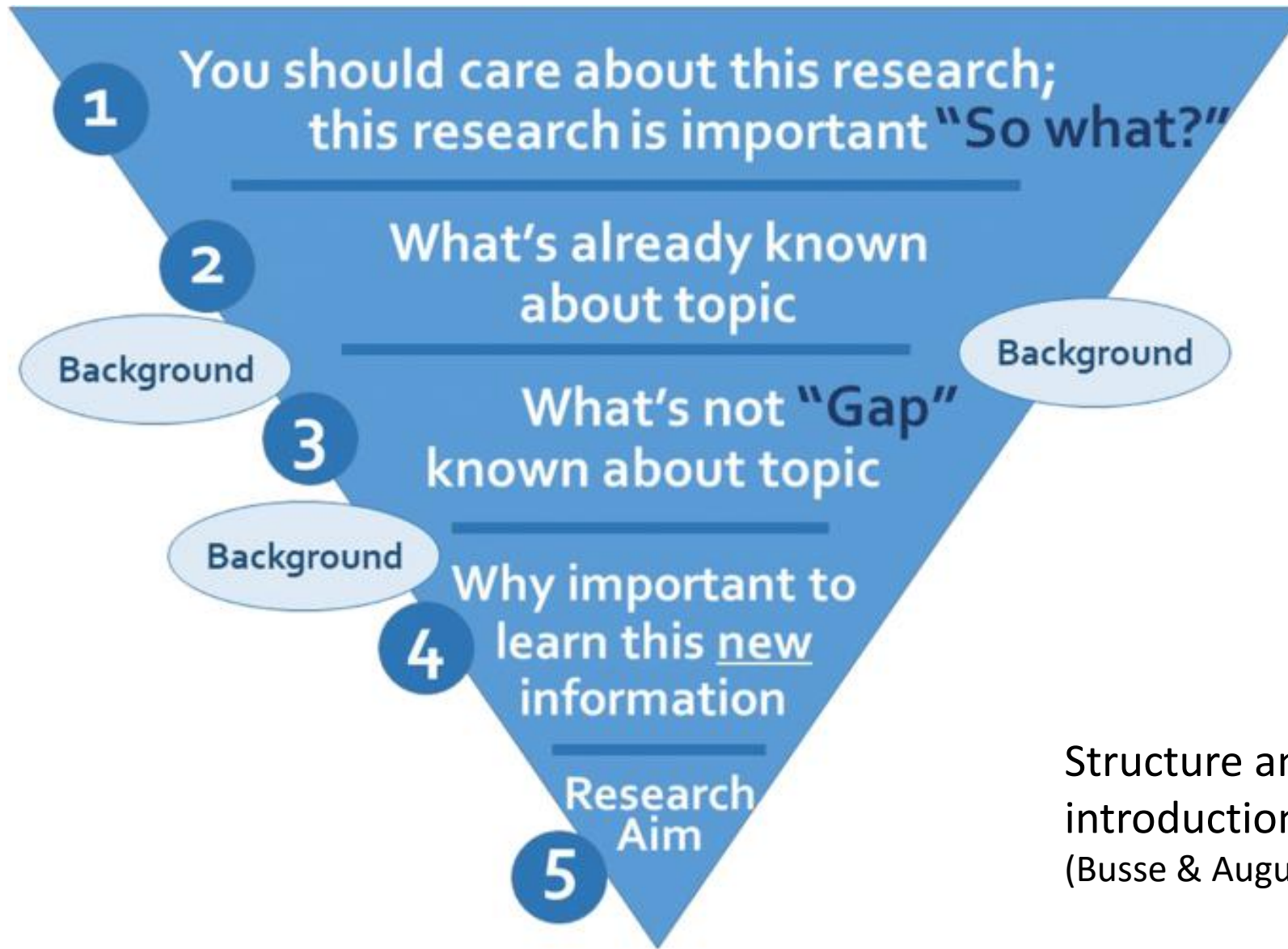
- Present evidence
- Justify methodology
- Comment on the data

Move 3: Summarize the data

Specific



General



Structure and content of an
introduction section
(Busse & August 2020)

Contents of the Introduction

A good introduction takes your reader from a familiar starting point and guides them through 5 stages:

1. sparking interest in the general problem or issue of concern
2. explaining the state of current research about that issue
3. demonstrating a knowledge gap or problem that must be solved
4. offering your research question(s) or hypotheses,
5. summarizing your goal/solution to the problem (and possibly its value).

3 Moves in Introductions

(CARS Model,
John Swales)



1. Establish a territory

- Step 1 Claiming centrality and/or
- Step 2 Making topic generalizations and/or
- Step 3 Reviewing items of previous research

2: Establish a niche

- a) Indicating a gap/problem, or
- b) making a counter-claim, or (c) raising a question, or (d) continuing a tradition

3: Occupy the niche (Fill the gap)

- Step 1 (a) Outlining aim/purposes, and/or (b) Announcing present research
- Step 2 (possible) Announcing principal findings
- Step 3 Indicating Research article/thesis structure

Differences between a study background and a literature review (Refer to Lesson 3)

- Study background relates to first three steps in Move 1 on the previous slide. It is short and concise. It highlights the significance of the study; see next slide 'centrality claim'.
- Literature review usually comes after background and introduction section. It is lengthy and detailed; it evaluates the progress of knowledge in the research area.

The opening sentence and centrality claim in the Introduction



- Captivate the reader's attention
- Show that the general research area is important, central, interesting, problematic or relevant in some way, that the study is worth reading and valuable.

Task: Examine some articles/theses in your field:

- What kind of opening sentence is used?
- How does the writer claim centrality for his/her research?
- Which words are used to refer to the importance of the research? Write them down.
- Compare the words you have found with words used by other writers in your fellow students' texts.

Introduction: some types of opening sentence in the Introduction

Type of sentence	Examples
General statement	1. <i>Coral bleaching is spreading because of rising sea temperatures.</i> 2. <i>Neglected tropical diseases are a group of 13 major disabling conditions that are among the most common chronic infections in the world's poorest people.</i>
Statistics	<i>Malaria is a major global health threat that was linked to an estimated 229 million cases of malaria and 409 000 deaths in 2019.</i>
Problem/Gap statement	<i>Despite massive research efforts, no Ebola vaccine exists to date.</i>

Centrality Claim

Reviewing previous research

- Citations: the opening paragraph likely refers to **reviews**, while the following paragraphs rather refer **to primary papers**
- Perceived value and credibility of citations:

Stance	Tense	Example
Personal (not for the 1st sentence)	past tense	Jones et al. reported that x affects blood sugar levels (Jones et al.).
Impersonal	present perfect (here passive)	Blood sugar levels have been shown to be affected by x (Jones et al.).
Factual	present tense	Blood sugar levels are affected by x (Jones et al.).

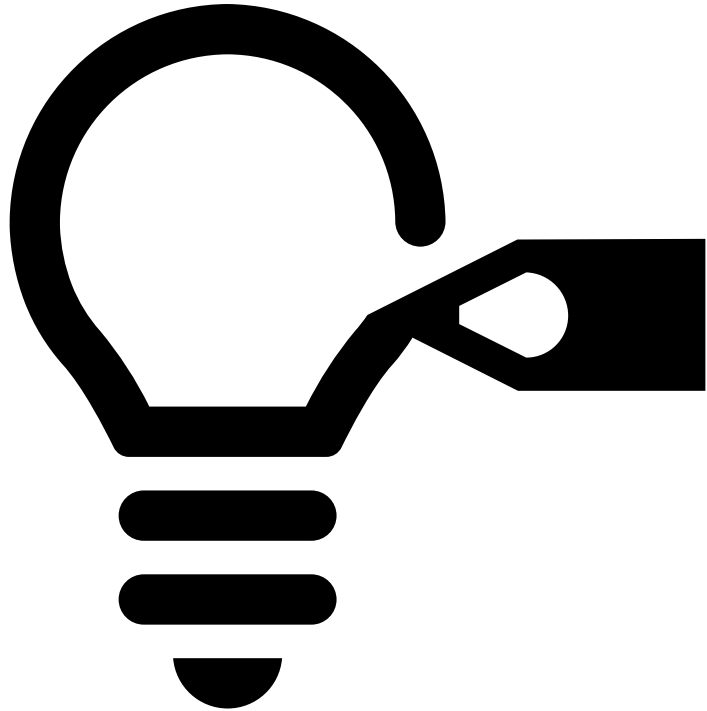
Some phrases often used in the centrality claim

- X is a key element of ...
- X is a widespread disease in ...
- X is a common disorder characterised by...
- ...there has been growing interest in...
- Recent studies have focused on...
- ... has become a major issue...
- ... remains a serious problem..
- ... there has been increasing concern...
- ...has been investigated by many researchers...
- ...has become an important aspect of...

Many more phrases in the script.

Refer to the Academic Phrasebank for more ideas

<http://www.phrasebank.manchester.ac.uk/>



Task 2: The opening sentence and centrality claim in the Introduction

Task: Examine some articles/theses in your field, and/or in your own article(s):

- What kind of opening sentence is used? Identify the type of opening sentence according to the previous slides.
- Do you think this choice was effective for this paper, or would a different type of sentence have been more effective?

Establishing a niche / Writing a gap/problem statement (Move 2): crucial part

- Indicates that your research is valuable by pointing out a problem or gap in previous research, or by extending previous research in some way
- States the motivation for doing your study and criticizes the fact that something important has been omitted from research so far.

Examples of gap sentences:

- ‘However, little is known about students’ views of the use of tablet PCs in class.’

Or

- ‘Apart from a small study by Cooper (2012), this phenomenon has not been the focus of research so far.’

Gap/Niche activity



CAN YOU FIND A SENTENCE OR PARAGRAPH THAT DEFINES THE GAP IN LITERATURE THAT ESTABLISHES THE NICHE FOR THE RESEARCH UNDERTAKEN IN THIS STUDY? WRITE IT DOWN.

COMPARE THIS STATEMENT TO THOSE USED IN OTHER STUDENTS' TEXTS.

WRITE DOWN WAYS IN WHICH ONE CAN INDICATE THAT A GAP EXISTS IN THE LITERATURE.

Statements to identify the gap

Quasi-negative statements are frequently used, including words such as ‘little’, ‘few’, ‘inadequate’, ‘no/none’:

- However, little information.. (not enough), sparse/scant information
- Little work.... Little data... Little research...
- Few studies/investigations... Few researchers... Few attempts...
- No studies/calculations/data to date have...
- None of these studies/calculations have...

(Be sure of your facts, else add ‘To the best of our/my knowledge, no studies...’)

Refer to the **Academic Phrasebank** for more examples.

Language notes

- ‘Little’ is used with uncountable nouns (‘Little research’), whereas
- ‘few’ is used with countable nouns (‘Few studies’). They both mean ‘not enough’.
- But: ‘a few’ or ‘a little’ means ‘some’

For much more input about scientific writing and more examples, see section 3 ‘Additional resources’ on ADAM.



Quasi-Negative verbs & adjectives to establish the gap

VERBS

Disregarded
Failed to consider
Concentrated on
Been limited to
Neglected to consider
Overestimated
Underestimated
Overlooked
Been restricted to
Lacked
Misinterpreted
Ignored

ADJECTIVES

Controversial
Incomplete
Inconclusive
Unconvincing
Unsatisfactory
Ambivalent

(Swales & Feak,
2012:351)

Examples of gap statements from the Academic Phrasebank (note frequent use of 'however')

However, very **little is known** about X in...
However, **few studies** have investigated ...
However, the nature of X **remains unclear**.
However, much **less is known** about how ...
However, the use of X **has not been investigated**.
Previous studies of X **have not dealt with** ...
Researchers have not treated X in much detail.
Such explanations are unsatisfactory because they ...
Most studies in the field of X have only focused on ...
Such approaches, however, have failed to address ...
Previous published studies are limited to local surveys.
Half of the studies evaluated failed to specify whether ...

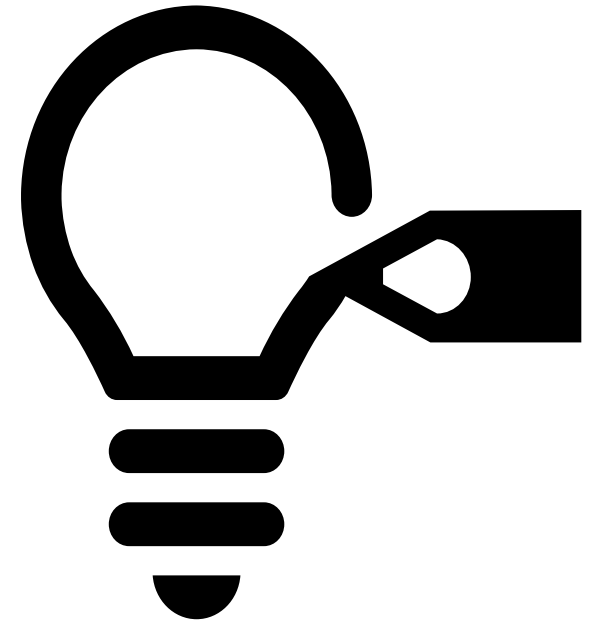
There is little published data on ...
No previous study has investigated X.

The use of X has **not** been investigated.
There has been **no detailed investigation** of ...
There has been **little quantitative analysis** of ...
Data about the efficacy and safety of X are **limited**.
Up to now, far **too little** attention has been paid to ...
A search of the literature revealed **few studies** which ...

Task: phrases used in gap/problem statement

Examine the Introductions of the articles you brought with you.

1. Identify the centrality claim and gap/ problem statement.
2. Were any of the above or similar phrases used? Write them down.
3. **Now, write your own gap statement, playing around with some of the above ways of doing that.**



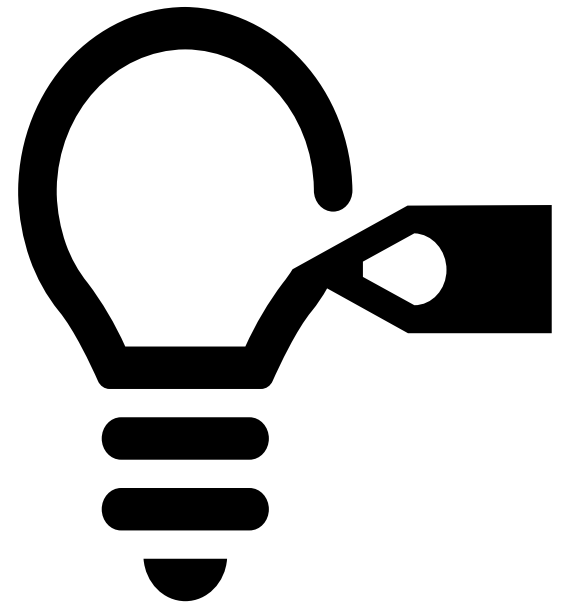
Purpose
statement to fill
the gap/
occupy the
niche/
address the
problem

- Authors indicate the aim, main purpose or main feature of the research.
- Mostly refer to the **type of text** or genre, such as paper, article, dissertation, thesis, report, etc., or
- to the **type of investigation**, such as experiment, investigation, study, survey, etc.
- Increasingly: 'We'...

Task: Purpose statement/ Filling the gap / addressing the problem

Examine the articles or theses that you brought to class with you.

- Can you find a sentence or paragraph that makes a statement about the purpose of the research: how the research aims to fill the gap in literature? Write it down.
- Compare this statement to those used in other articles. Write down different ways that are used to describe how the gap will be filled by your research.
- Is the first person singular (I) used? And the first person plural (We)?



Some
statements to
describe the
purpose of the
study
(to fill the
research gap):
Move 3

- The aim of this research is to give...
- This study reports on the results obtained..
- The main purpose of the experiment reported here was to...
- This study was designed to evaluate...
- The present work extends the last model...
- The primary focus of this dissertation is..
- The aim of this investigation is to...
- The primary objective of this paper is to provide...

(Adapted from Swales and Feak, 2012:355)

More phrases in script and on Academic Phrasebank

Filling the gap/Occupying the niche

Writers could fill the gap / occupy the niche by stating their research hypothesis or research questions.



Hypothesis

A hypothesis provides a tentative answer to your research question that has not yet been tested. It includes a proposed answer to a question/problem that can be verified by testing – a prediction of the anticipated outcome of the study.

For some research projects, you might have to write several hypotheses that address different aspects of your research question.

A hypothesis is an educated guess about the relationship between factors – it should be based on existing theories and knowledge. It also has to be testable, which means you can support or refute it through scientific research methods (such as experiments, observations and statistical analysis).

Variables in hypotheses

- Hypotheses propose a relationship between two or more types of variables.
- An [independent variable](#) is something the researcher changes or controls.
- A [dependent variable](#) is something the researcher observes and measures.
- If there are any [control variables](#), [extraneous variables](#), or [confounding variables](#), be sure to jot those down as you go to minimize the chances that [research bias](#) will affect your results.

(See <https://www.scribbr.com/methodology/hypothesis/>)

Hypothesis: examples here and in script

You can use an 'If... then' format:

- **If** skin cancer is **related** to ultraviolet light, **then** people with a high exposure to ultra-violet light will have a higher frequency of skin cancer.
- **If** leaf colour change is **related** to temperature, **then** exposing plants to low temperatures will result in changes in leaf colour.

You can express the **correlation between variables**:

- The number of lectures attended by first-year students has a positive effect on their exam scores.

If you are **comparing two groups**, the hypothesis can state what **difference you expect to find** between them.

- First-year students who attended most lectures will have better exam scores than those who attended few lectures.

(See <https://www.scribbr.com/methodology/hypothesis/>)

Research questions

- Mostly 'what' or 'how' questions
- Open-ended (not yes/no answers)
- Describe rather than compare or relate variables
- Questions can be direct or indirect
(See examples in script.)

Consider the papers that you brought with you. Do they contain research questions, maybe in an indirect question format?

How many? Direct or indirect?

Formulating research questions:

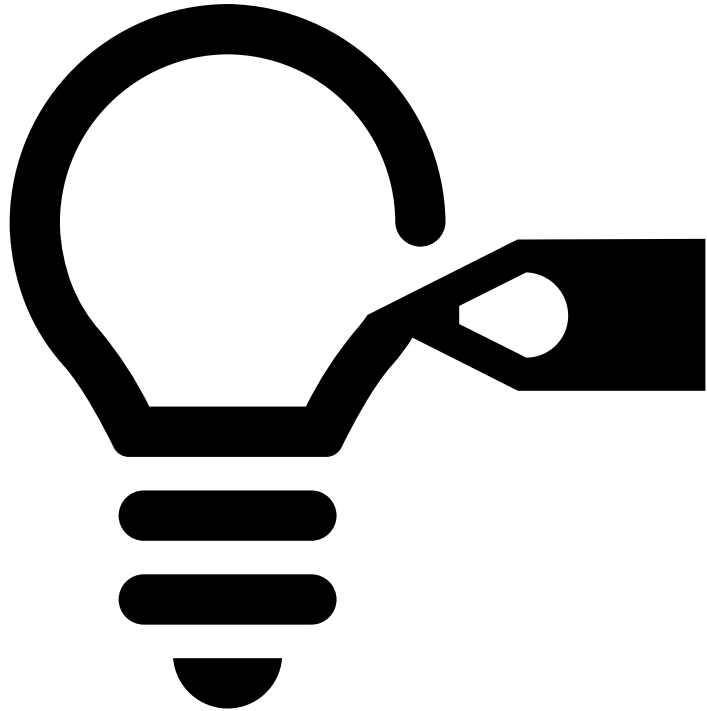
Use open-ended, researchable questions

Research question	Explanation
✗ Does the US or the UK have a better healthcare system? ✓ How do the US and the UK compare in health outcomes among low-income people with chronic illnesses?	The first question is too broad and subjective: there's no clear criteria for what counts as "better." It is also not open ended. The second question is much more researchable. It uses clearly defined terms and narrows its focus to a specific population.
✗ Does gaming have an effect on the mind? ✓ What effect does 2 hours of daily gaming have on the attention span of 12-16 year-olds?	The first question is not specific enough. It is also not open ended. The second question is more researchable using qualitative and quantitative data collection.

Direct or indirect research questions

- **Direct:** How does the use of insecticide-treated bed nets (ITNs) affect malaria illness, severe disease, and death due to malaria in endemic regions?
- **Indirect:** This study investigates whether insecticide-treated bed nets (ITNs) affect malaria illness, severe disease, and death due to malaria in endemic regions.

(Note word order in indirect questions – same as in statements.)



Task: hypothesis / research question(s)

Analyse the introduction section of the thesis that you brought with you.

1. Does the thesis contain a hypothesis? If so, how is the hypothesis formulated? Write it down.
2. Does the hypothesis include a prediction of the expected outcome, and is it testable? Compare your findings with your fellow students.
3. If there is no hypothesis, are there any **research questions**? If so, how are the research questions formulated?
4. Did the writer perhaps use a hypothesis as well as research questions?



Task: Formulate
a hypothesis
and a research
question for
your study

- Now write hypotheses or research questions relating to your field of study/your study topic.
- Formulate the problem that you want to investigate in your study as a hypothesis and then as a research question.
- Which of the formats do you prefer? Why?

Expected impact/implications of the study

- At the end of the introduction
- Don't overstate claims of the study's significance.

Examples:

- This study provides new insights into...
- The findings should make a valuable contribution to the field of
- Characterisation of X is important for our increased understanding of ...
- The findings could aid decision-making for healthcare providers..
- This study aims to contribute to this growing area of research by exploring ...

Limitations and structure

- Some reference to limitations of the study might be made. Example: *It is beyond the scope of this study to examine the ...*

The author might also refer to the structure of the manuscript, more frequently in a thesis or research report than in an article. Examples:

- *The overall structure of the study takes the form of six chapters, including ...*
- *Chapter Four analyses the results of interviews and focus group discussions undertaken during ...*
- *Chapter Two begins by laying out the theoretical dimensions of the research, and looks at how ...*
- *The fourth section presents the findings of the research, focusing on the three key themes that ...*

Using participle clauses for more concise expression or to join sentences

Participle clauses enable us to state information in a more economical, sometimes more elegant way. They are used in written academic/scientific, journalistic and literary texts.

They are formed using

- present participles (*going, reading, seeing, walking, etc.*),
- past participles (*gone, read, seen, walked, etc.*) or
- perfect participles (*having gone, having read, having seen, having walked, etc.*).

For example, instead of saying:

- ‘While we distributed the mosquito nets, we also analysed the structure and building materials of the dwellings.’

You could say:

‘**Distributing** the mosquito nets, we also analysed the structure and building materials of the dwellings.’

Present participle clauses

Here are some common ways we use present participle clauses. Note that present participles have a similar meaning to active verbs.

- To give the result of an action
*The bomb exploded, **destroying** the building.*
- To give the reason for an action
***Knowing** she loved reading, Richard bought her a book.*
- To talk about an action that happened at the same time as another action
***Standing in the queue**, I realised I didn't have any money.*
- To add information about the subject of the main clause
***Starting in the new year**, the new policy bans cars in the city centre*

Past participle clauses

Here are some common ways that we use past participle clauses. Note that past participles normally have a passive meaning.

- With a similar meaning to an *if* condition, so instead of saying '*If you use participles in this way, they can make your writing more concise.*', you can say **Used in this way**, participles can make your writing more concise.
- To give the reason for an action **Worried by the news**, she called the hospital.
- To add information about the subject of the main clause **Filled with pride**, he walked towards the stage.

Perfect participle clauses

Perfect participle clauses show that the action they describe was finished before the action in the main clause. Perfect participles can be structured to make an active or passive meaning.

So, instead of saying 'After he had collected the samples, he took them to the lab for analysis.' you can say

- **Having collected** *the samples, he took them to the lab for analysis.*
- **Having finished** *their analyses, they will compare their findings to other studies. (Instead of 'After they have finished their analysis, they will compare...')*
- **Having been awarded** *their degrees, they started looking for new jobs.*

Participle clauses after linking words and prepositions

It is also common for participle clauses, especially with *-ing*, to follow conjunctions (linking words) and prepositions such as *before*, *after*, *instead of*, *on*, *since*, *when*, *while* and *in spite of*.

- ***Before cooking***, you should wash your hands.
- ***Instead of complaining about it***, they should try doing something positive.
- ***On arriving at the hotel***, he went to get changed.
- ***While packing her things***, she thought about the last two years.
- ***In spite of having read the instructions twice***, I still couldn't understand how to use it.

For much more input about scientific writing and more examples, see section 3 'Additional resources' on ADAM.

Assignment 4: Write an Introduction section

- Write 3 different versions of possible introductory sentences for your thesis/ proposal/ article.
- Write a 250-350-word **introduction section** to the manuscript (e.g., proposal, article, thesis) you plan to work on for the remainder of the course. Include the following elements, **highlighted** in **different colours** in your text, and **identified** in the margin.
- A suitable introduction statement
- A very brief summary of or reference to some relevant literature
- A statement defining the problem or gap in the research/literature
- Your research hypothesis or research questions, or the study goal(s).
- A statement or paragraph in which you show how you intend to fill the gap/niche in research.
- Reference all sources used in the text and in a short bibliography.
- Review a colleague's introduction section.
- Put your paragraph through the plagiarism check, Turnitin, on the Turnitin ADAM page. Upload the Turnitin report too (not the receipt).